Tetrahedron User Guide

Nov 9, 2026 · Tetrahedron developers

Overview

Tetrahedron shows the results of finite element analysis (FEM) on 3D and 2D models. You load a mesh, then color it by stress, deformation, or any other value you computed elsewhere. It can also build a mesh for you from a CAD file.

The app supports one 3D element type, the four-node tetrahedron, which gives the app its name. 2D models use three-node triangles.

The home screen has two modes:

Mode	Home screen card	What it does
Model viewer	Show model and calculation results	Loads a mesh from text files and colors it by stress, displacement, or a custom value
Mesh generator	Create finite element mesh	Turns an .stl or .step file into a tetrahedral (or triangular) mesh and saves it as files the viewer can load

Image — Home screen with the two mode cards

The interface is available in English, Ukrainian, German, and Polish. The app picks your system language at launch, and you can switch it from the language menu in the top bar.

Installation

Install Tetrahedron from the installer for your system. It ships with everything it needs, so you don't have to install Java separately.

- **Windows:** run the .msi installer and follow the steps. Start Tetrahedron from the Start menu.
- **macOS (Apple Silicon):** open the .dmg and drag Tetrahedron into Applications. The app is signed and notarized, so it opens without security warnings.
- **From source:** install JDK 21 and Maven, run `mvn package`, then run `java -jar target/tetrahedron-desktop-<VERSION>-jar-with-dependencies.jar`.

The **?** button in the top bar opens the online instructions on the project wiki.

The interface

Both work modes use the same layout: a top bar, a tool strip on the left, the 3D view in the middle, and a settings panel on the right.

Image — Model viewer: left toolbar, legend, a layered model colored by Mises stress, th…

Top bar: click the Tetrahedron logo to go back to the home screen. Use the language menu to switch languages, and the **?** button to open the online help.

Left toolbar

Button	What it does
Cursor	Drag to rotate the model freely. This is the default.
Up–down arrows	Dragging rotates the model only around the horizontal axis.
Left–right arrows	Dragging rotates the model only around the vertical axis.
Palette	Sets the model and background colors.
Legend (viewer only)	Sets the number of color bands, their value ranges, and the color theme.
Trash (viewer only)	Closes the current model and opens the file picker to load another one.

Mouse controls in the 3D view

- **Drag** rotates the model, using the mode picked in the left toolbar.
- **Scroll** zooms in and out. Scrolling with Shift held down doesn't zoom.
- **Click** a point on the model to show the nearest node's number and its X, Y, Z coordinates in the info panel.

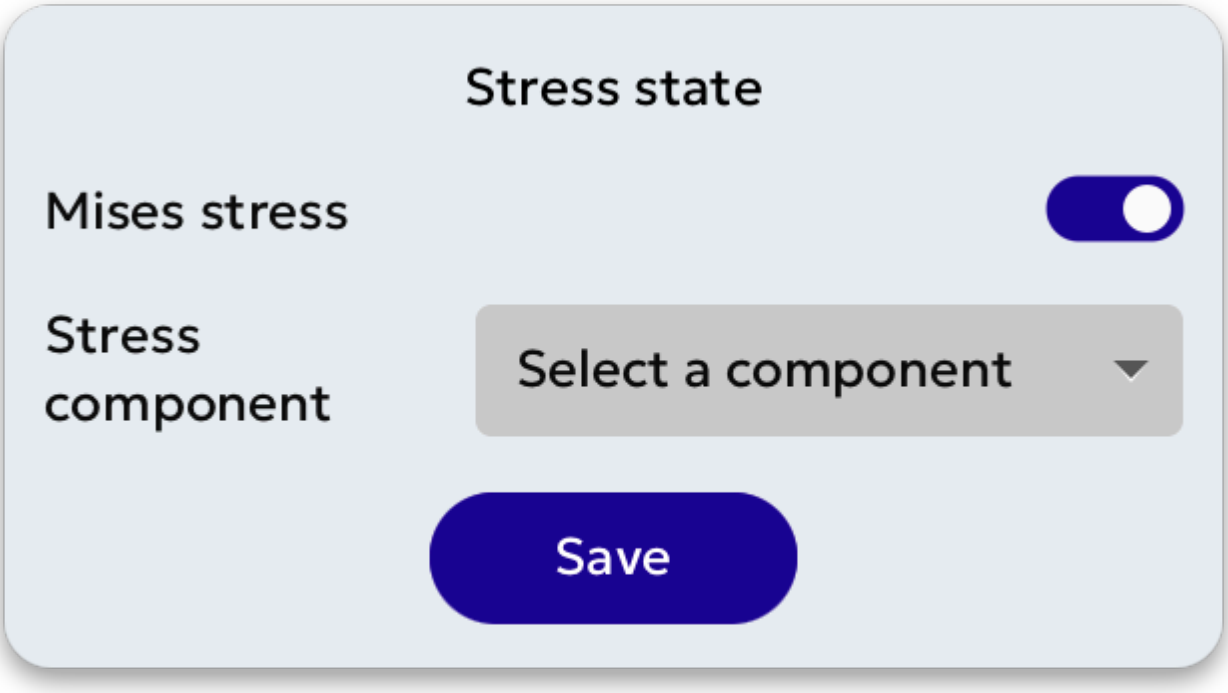
Display switches (top of the right panel)

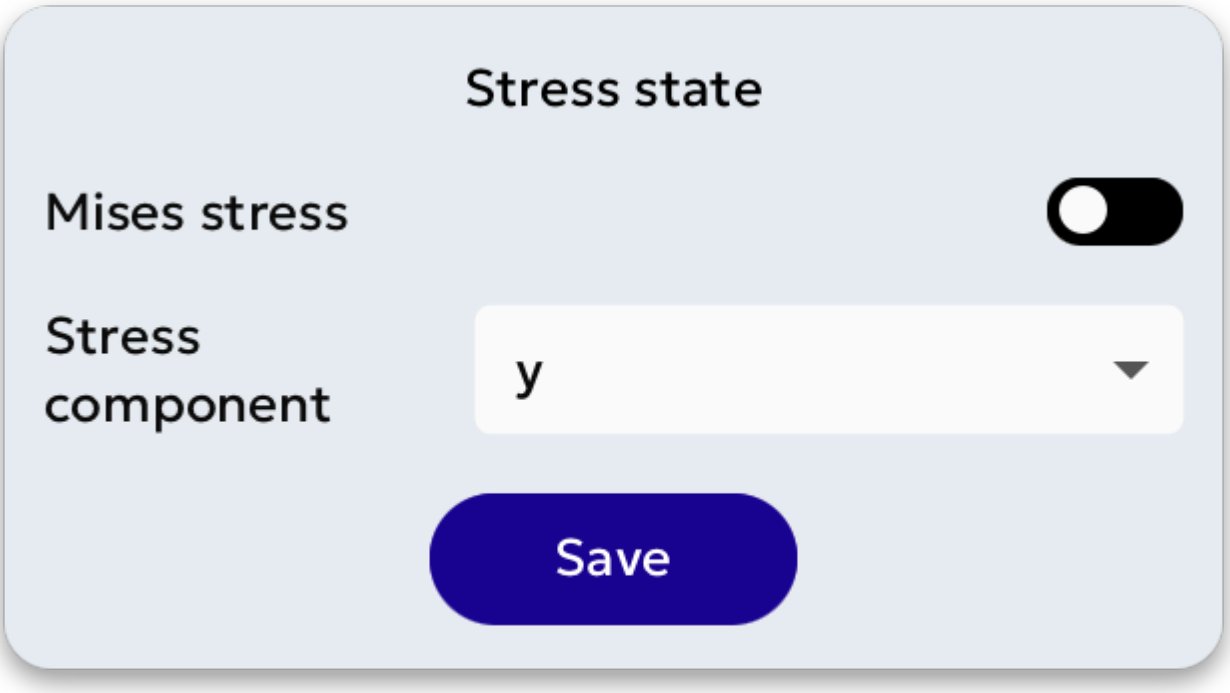
- **Elements mesh** shows or hides the element edges.
- **Light** turns shading on or off. Shading makes the model's shape easier to see, while flat colors are easier to read against the legend.

Viewing a model and its results

1. On the home screen, click **Start** under *Show model and calculation results*.
2. In the dialog, set **Model dimension** to 3D or 2D.
3. Pick the **coordinates matrix** file (the nodes) and the **indices matrix** file (the elements). See *File formats* below.
4. Click **Create model**. If the coordinates file has nodes that no element uses, the app warns you and asks whether to go on.
5. Use the right panel to load result data onto the model. Only one data set is colored at a time, and the legend updates to match it.

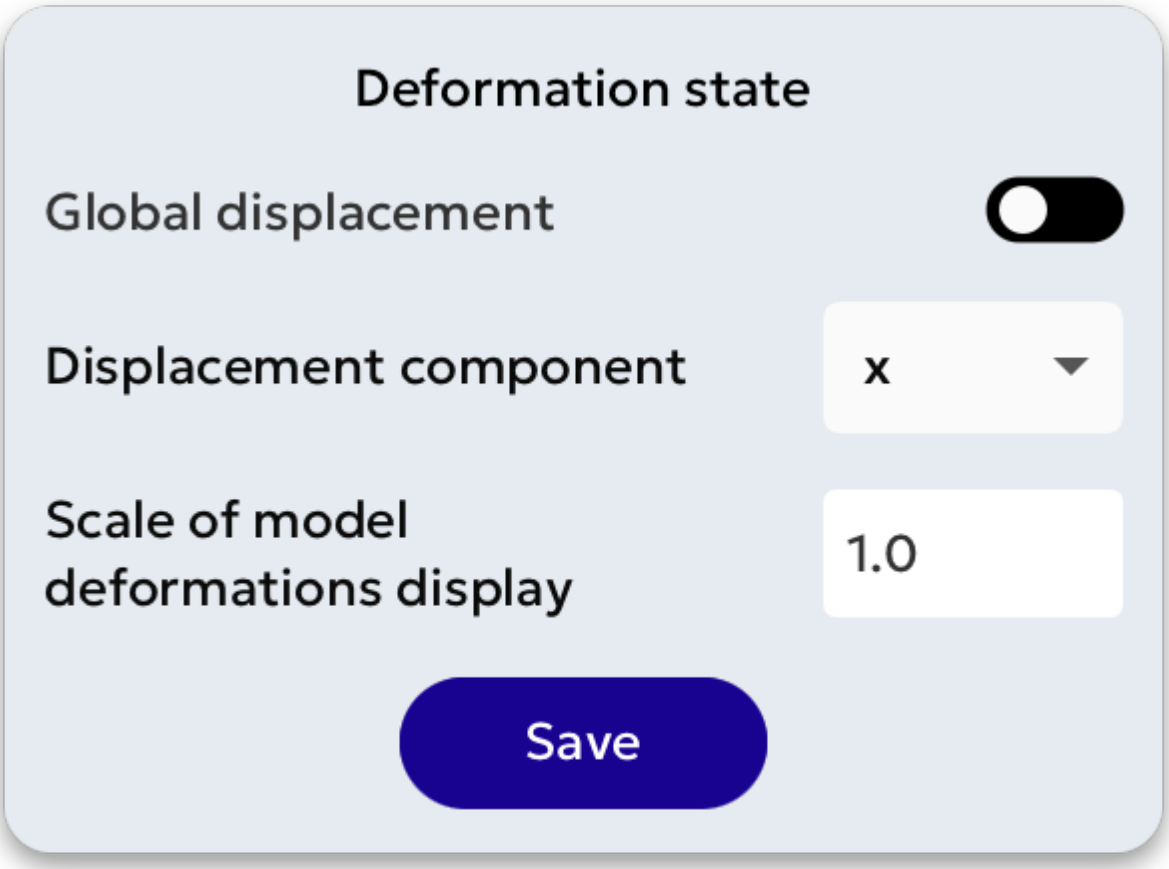
Stress state. Click **Load file** and pick a stress file. By default the model is colored by von Mises stress. To show a single component instead (x, y, z, xy, yz, xz), click the settings button next to the heading. Turn off *Mises stress* there, pick the component, and click **Save**.





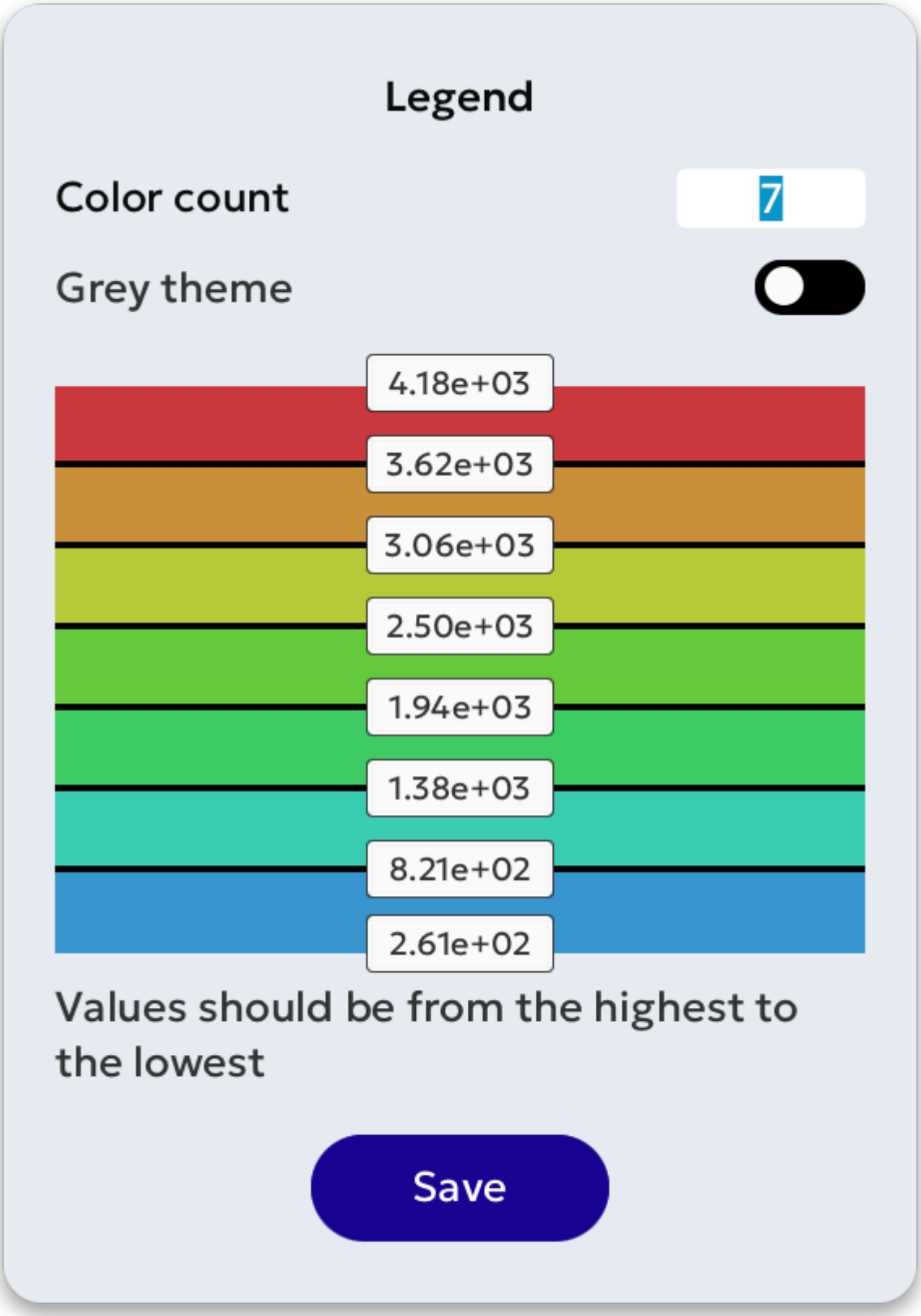
Deformation state. Click **Load file** and pick a displacement file. The model is drawn in its deformed shape and colored by total displacement. In the settings dialog you can:

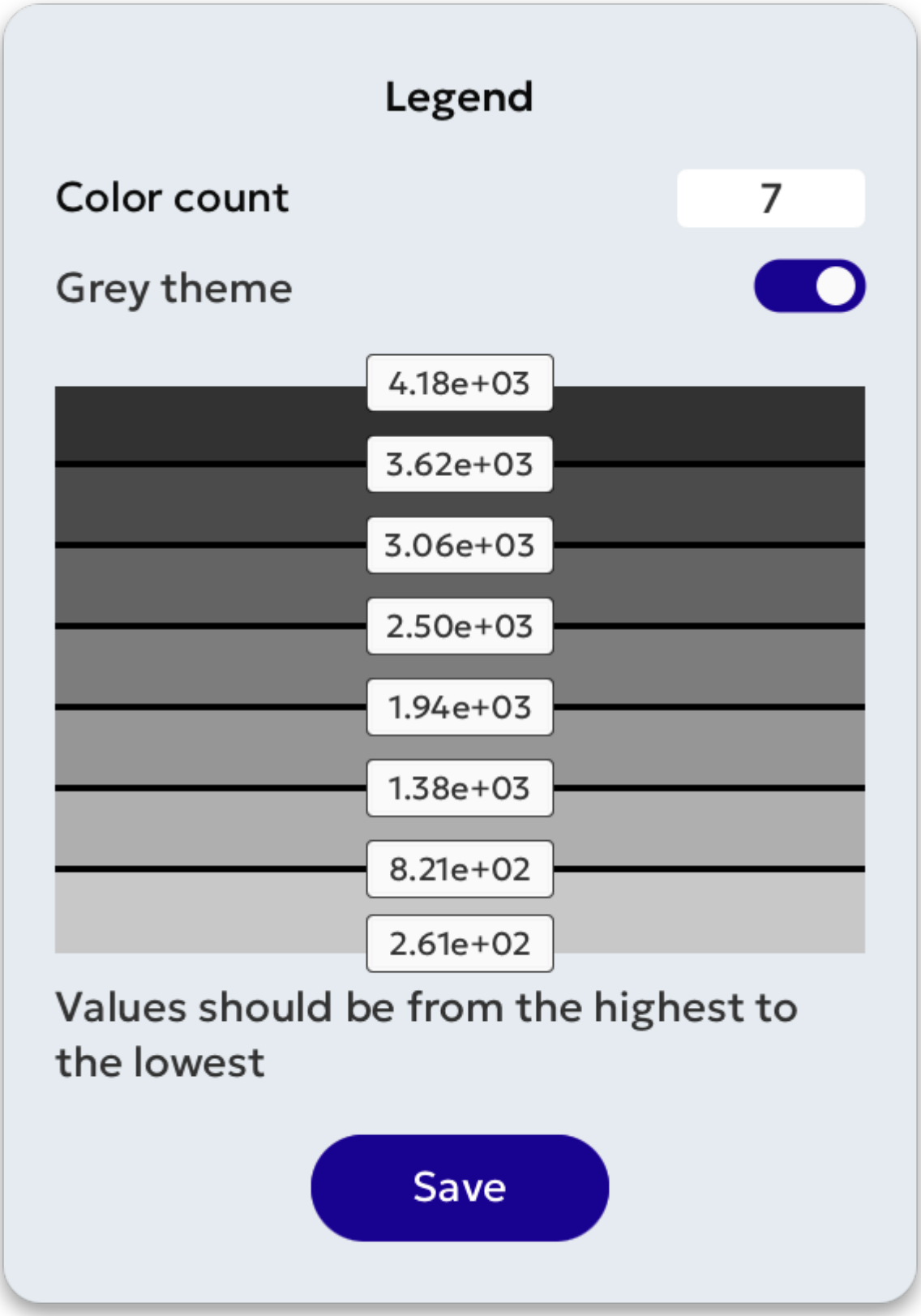
- turn off *Global displacement* to color by a single component (x, y, or z)
- change the **scale of model deformations display**, for example 100 to make small deformations visible (default 1.0)



Other characteristic. Use this for any single value per element, such as temperature, strain energy, or a safety factor. Click **Load file** and pick the file.

Legend. Click the legend button in the left toolbar to set the **color count** (2 to 15) and edit each band's range. List the ranges from highest to lowest. Turn on **Grey theme** for a grayscale legend instead of the rainbow one, which works better for printing.


Legend
Color count
7
Grey theme
4.18e+03
3.62e+03
3.06e+03
2.50e+03
1.94e+03
1.38e+03
8.21e+02
2.61e+02
Values should be from the highest to the lowest
Save


Legend
Color count
7
Grey theme
4.18e+03
3.62e+03
3.06e+03
2.50e+03
1.94e+03
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Values should be from the highest to the lowest
Save

Generating a mesh

The mesh generator turns a CAD file into a finite element mesh using the Gmsh library. It saves the mesh in the same text format the viewer reads.

1. On the home screen, click **Start** under *Create finite element mesh*.
2. Set **Model dimension** to 3D or 2D, pick an .stl or .step file, and click **Open model**. The app shows the model's surface.
3. Check the mesh settings in the right panel. The app fills in the minimum and maximum sizes from the model's dimensions.
4. Click **Mesh**. A progress dialog shows the status (In progress, Success, or Failure) and the Gmsh log. When it finishes, the panel shows the node and element counts.
5. Click **Save model** and choose a folder.

Image — Mesh generator after opening an STL file: mesh sizes filled in, Save model not …

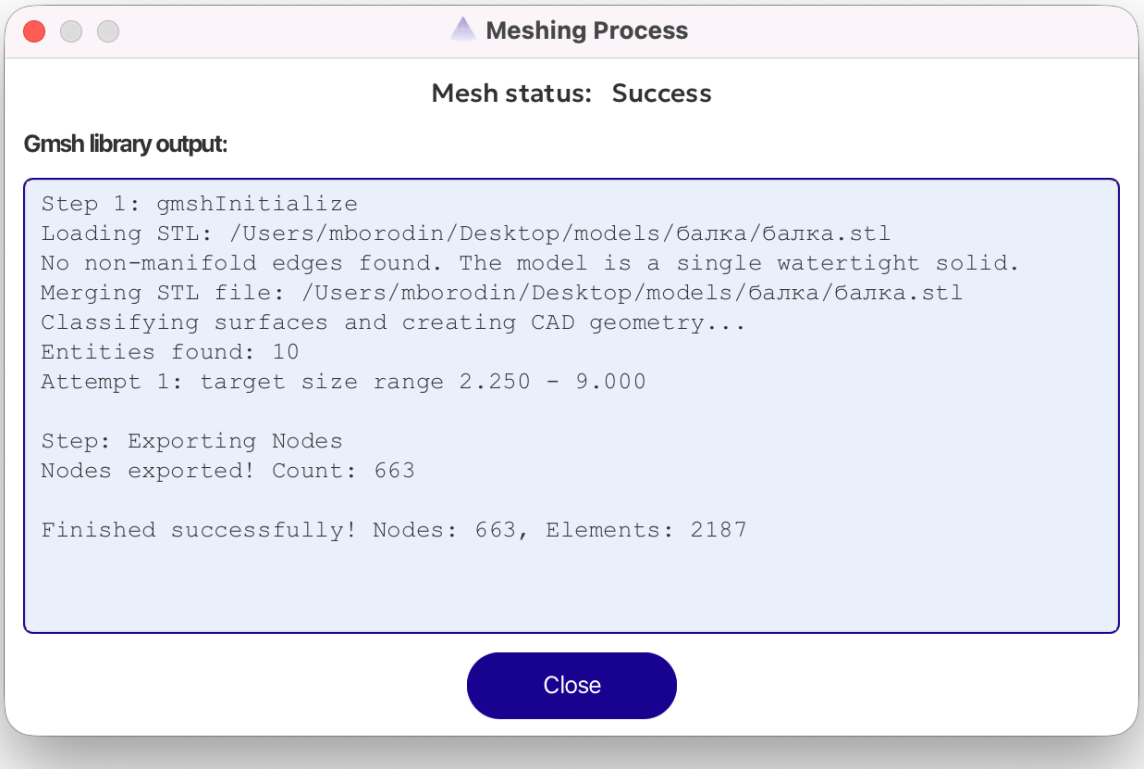


Image — The meshed model with 1,130 nodes and 3,753 elements (min size 0.33, max size 3…

Setting	Meaning	Default
Min mesh size	Smallest element edge length, in model units	Calculated from the model
Max mesh size	Largest element edge length, in model units	Calculated from the model
Angle to define surfaces	Angle in degrees between facets above which an STL edge is treated as a sharp edge between two surfaces	40

Smaller sizes give a finer, more accurate mesh, but meshing takes longer and produces bigger files.

Which input file to use

- **.stl:** a single part. For 3D, the surface must be closed. 2D models are flat outlines. All elements go into layer 1.
- **.step / .stp (3D only):** multi-body models, such as stacked layers or assemblies. Where parts touch, the app merges their shared boundaries, so the nodes match exactly across layers. Each element is tagged with the number of the body it belongs to, so a solver can give each layer its own material.

Click **Change model** to load a different file without leaving the page.

File formats

All data files are plain text with one record per line. Values are separated by spaces or tabs, and numbers can use scientific notation (`5.857E+10`). In every file, the leading index column is optional. Node and element numbers start at 1.

File	One line per	Columns (3D)	Columns (2D)
Coordinates matrix	node	[index] x y z	[index] x y
Indices matrix	element	[index] n1 n2 n3 n4	[index] n1 n2 n3
Stress	element	[index] x xy zx y yz z	—
Displacement	node	[index] dx dy dz	—
Other characteristic	element	[index] value	[index] value

The displacement file must have exactly one line per node. Every node number in the indices matrix must exist in the coordinates matrix.

Example of a coordinates matrix, followed by an indices matrix:

```
 1 0 1 0
 2 0 0 0
 3 1 1 0
```

```
1 12 14 10 15
2 12 14 5 3
```

Files written by the mesh generator

File	Contents
0_constants.txt	Nodes per element, node count, and element count
1_coordinates_matrix.txt	Node coordinates, renumbered from 1
2_indices_matrix.txt	Element number, starting at 1, followed by its node numbers
3_layer_ids.txt	Element number and its layer (body) number, starting at 1

You can open 1_coordinates_matrix.txt and 2_indices_matrix.txt straight away in the model viewer.

Troubleshooting

When a file can't be read, the app shows an error dialog that names the problem and often the line that caused it. Fix the file, then load it again.

Message	Cause and fix
There should be 3 coordinates for each node	The coordinates file has the wrong number of columns. Check that Model dimension (2D or 3D) matches the file.
There should be 4 node indices for each element	The indices file has the wrong number of columns. 3D elements need 4 nodes, 2D elements need 3.

Message	Cause and fix
Indices matrix uses non-existing coordinates	An element refers to a node number that isn't in the coordinates file.
File contains repeating indices	The same index appears twice in the leading index column.
Displacement count is not the same as nodes count	The displacement file needs exactly one line per node.
Each line in file with stress data should have values for 6 stress components	The stress file needs 6 values per element. For a single value per element, load it as *Other characteristic*.
Legend color count must be between 2 and 15	Enter a color count from 2 to 15 in the legend settings.
Mesh generation failed after all attempts	Gmsh couldn't mesh the geometry. Check that the STL surface is closed, or try different min and max mesh sizes. The Gmsh log in the progress dialog shows details.
Not 2d mesh was loaded	A 2D model was requested, but the STL file isn't flat. Switch Model dimension to 3D, or use a flat file.

Tips

- Comma decimals are accepted in the mesh size fields (0,5 works like 0.5).
- If you can't see a deformation, increase the display scale in the Deformation state settings.
- To start over with a different model, click the trash button in the left toolbar.